



SIL Verification Report

Hybrit Pilot Plant

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	5/20/2020		Document created (First calculation of SIF)
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1 Purpose and Scope

This document, automatically generated by the exida exSILentia® software, details the results, selections, and assumptions made during the Safety Integrity Level verification for the Safety Instrumented System (SIS) and its associated Safety Instrumented Functions (SIF) for the Hybrit Pilot Plant project.

1.1 General Project Information

Project Identification: 001
Project Name: Hybrit Pilot Plant
Revision: 0
Company:
Project Description:

2 SIF-001 SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning

This chapter details the SIL Verification results, selections, and assumptions for the SIF-001 SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning Safety Instrumented Function.

2.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Low Pressure (254PI330 L1) nitrogen system, CO2 cleaning
SIF Tag	SIF-001
SIF Description:	254PI330 L1 low pressure on nitrogen close 254SV559 close 254SV561. Open bleed after GSS.

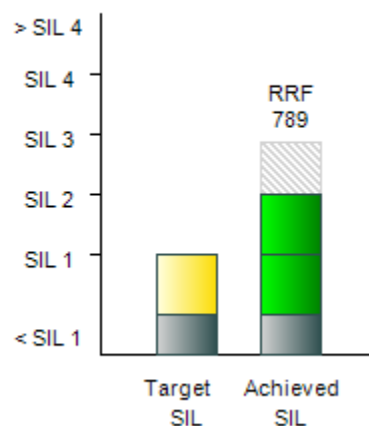
2.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 789.3



2.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-001 SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

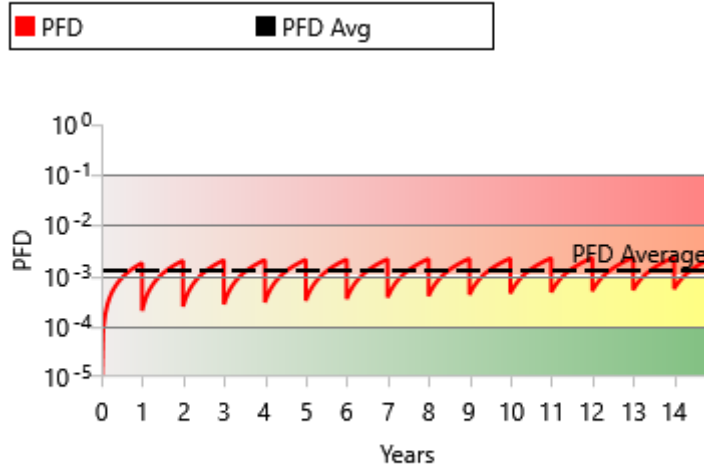
Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:
Design References
Comments:

Analysis Date: 10/4/2019

SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning

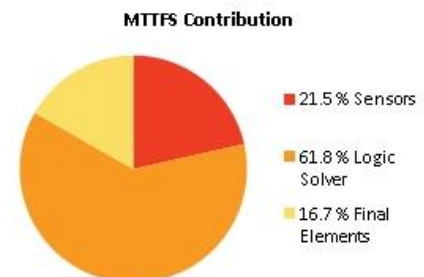
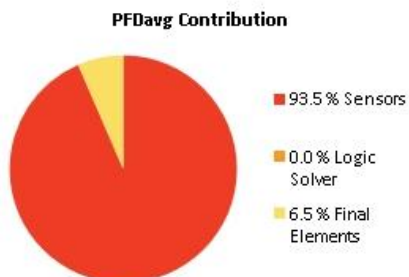


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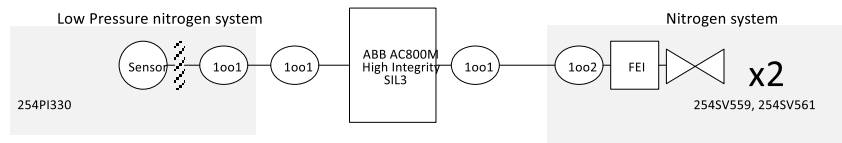
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-001 SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 1 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.27E-03	789.3	2	2	N/A	44.78



The SIF Conceptual Design is modeled as shown in the picture below.



2.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-001 SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning Safety Instrumented Function is quantified as follows.

Table 2 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.18E-03	2	N/A	208.68	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

2.3.1.1 Sensor Group 1: Low Pressure nitrogen system

The information and reliability data underneath describe the Low Pressure nitrogen system sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	254PI330	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75		2006.2.02	-	-	B	2
Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off						
PR Electronics 9106, 1 active in/1 active out		2014.4.03	-	-	A	2
ABB AC800M High Integrity SIL3 [AnalogIn]		2013.1.02	-	-	B	3
MRT [Hours]:	24					
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99			
Proof Test Execution:	Duration 1 hours					

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Low Pressure nitrogen system.

Table 3 Reliability Data Sensor Group Low Pressure nitrogen system

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
PR Electronics 9106, 1 active in/1 active out	1.46E-07 <i>0.00E+00</i>	2.70E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.46E-07</i>	0.00E+00 <i>0.00E+00</i>	4.10E-08 <i>6.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.77E-07 <i>1.77E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		73.4
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

2.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-001 SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning Safety Instrumented Function is quantified as follows.

Table 4 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β-factor [%]: 1

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 5 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		100.0

2.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-001 SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning Safety Instrumented Function is quantified as follows.

Table 6 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
8.25E-05	3	N/A	267.4	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

2.3.3.1 Final Element Group 1: Nitrogen system

The information and reliability data underneath describe the Nitrogen system final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg 254SV559,254SV56
(each): 1

	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Metso Norgren		-	-	A	3
Metso VPVL		-	-	A	3
Metso 7000 9000		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 85

Proof Test Execution:

Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Nitrogen system.

Table 7 Reliability Data Final Element Group Nitrogen system

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Metso Norgren		2.28E-09		2.28E-11			
Metso VPVL		4.25E-08		8.49E-09			
Metso 7000 9000		7.57E-08		4.84E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		79.8
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

3 SIF-002 SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system

This chapter details the SIL Verification results, selections, and assumptions for the SIF-002 SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system Safety Instrumented Function.

3.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system
SIF Tag	SIF-002
SIF Description:	Low differential pressure 254XI330 (254PI330/ 347PI306) protect nitrogen and close 254SV559 and close 254SV561

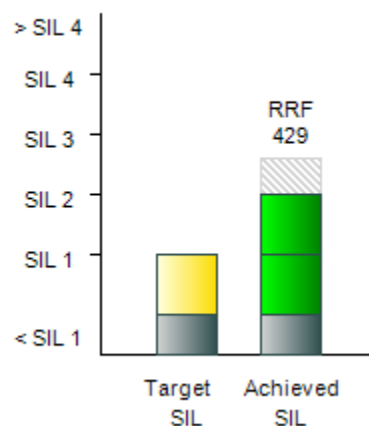
3.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 429.8



3.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-002 SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:

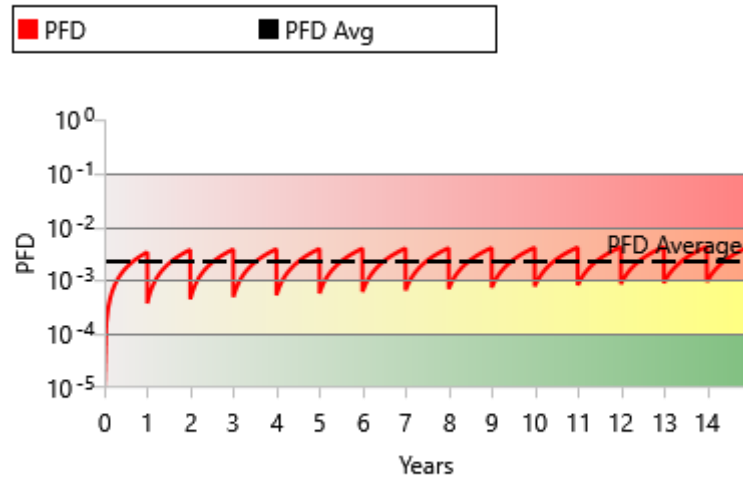
Analysis Date:

10/4/2019

Design References

Comments:

SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system

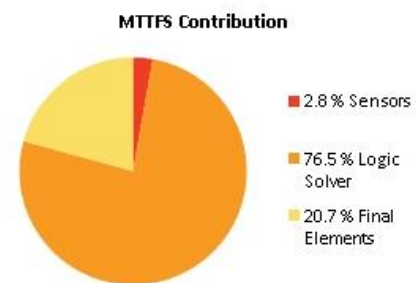
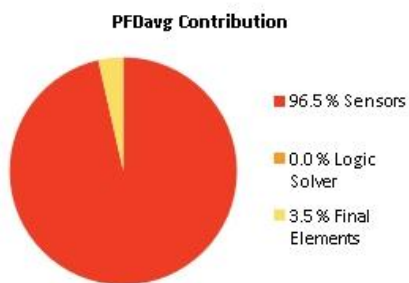


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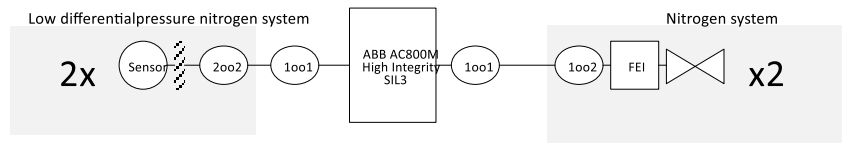
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-002 SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 8 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
2.33E-03	429.8	2	2	N/A	55.42



The SIF Conceptual Design is modeled as shown in the picture below.



3.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-002 SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system Safety Instrumented Function is quantified as follows.

Table 9 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.24E-03	2	N/A	1990.29	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

3.3.1.1 Sensor Group 1: Low differential pressure nitrogen system

The information and reliability data underneath describe the Low differential pressure nitrogen system sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo2 Voting type: Identical
HFT: 0 β-factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2
Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
PR Electronics 9106, 1 active in/1 active out	2014.4.03	-	-	A	2
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Low differentialpressure nitrogen system.

Table 10 Reliability Data Sensor Group Low differentialpressure nitrogen system

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
PR Electronics 9106, 1 active in/1 active out	1.46E-07 <i>0.00E+00</i>	2.70E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.46E-07</i>	0.00E+00 <i>0.00E+00</i>	4.10E-08 <i>6.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.77E-07 <i>1.77E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		73.4
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

3.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-002 SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system Safety Instrumented Function is quantified as follows.

Table 11 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β -factor [%]: 1

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 12 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

3.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-002 SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system Safety Instrumented Function is quantified as follows.

Table 13 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
8.25E-05	3	N/A	267.4	1	2

Number of Final Element group(s): 1
Voting between groups: 1001
β-factor [%]: N/A

3.3.3.1 Final Element Group 1: Nitrogen system

The information and reliability data underneath describe the Nitrogen system final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1002 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Metso Norgren		-	-	A	3
Metso VPVL		-	-	A	3
Metso 7000 9000		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 85

Proof Test Execution:

Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Nitrogen system.

Table 14 Reliability Data Final Element Group Nitrogen system

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Metso Norgren		2.28E-09		2.28E-11			
Metso VPVL		4.25E-08		8.49E-09			
Metso 7000 9000		7.57E-08		4.84E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		79.8
					ROUTE 2h COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

4 SIF-003 SIL1 High level (347LZ302) in Absorber (347YM100)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-003 SIL1 High level (347LZ302) in Absorber (347YM100) Safety Instrumented Function.

4.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 High level (347LZ302) in Absorber (347YM100)
SIF Tag	SIF-003
SIF Description:	High level (347LZ302 H1) in absorber (347YM100) close 347SV509 (USY feed water) and stop 346PU101 top gas compressor

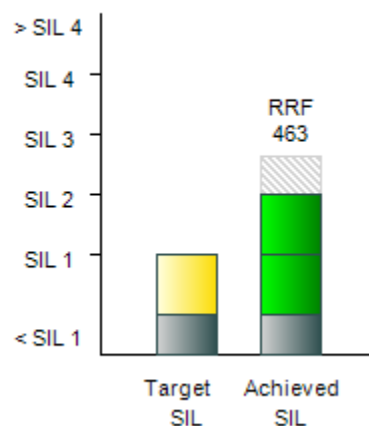
4.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 463.8



4.3 SIL Verification

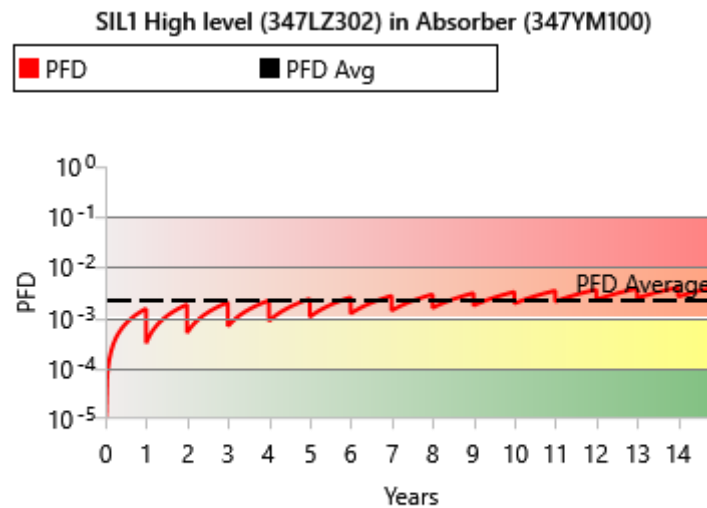
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-003 SIL1 High level (347LZ302) in Absorber (347YM100) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:
Design References
Comments:

Analysis Date: 10/4/2019

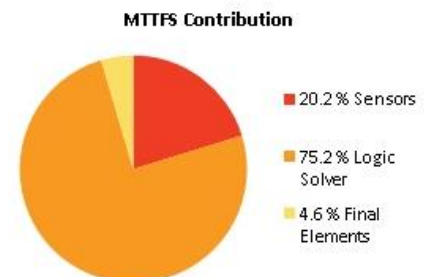
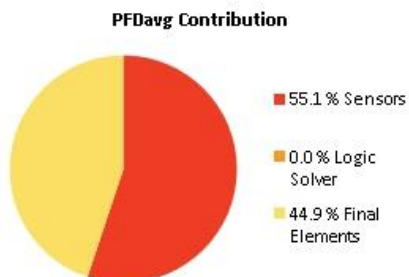


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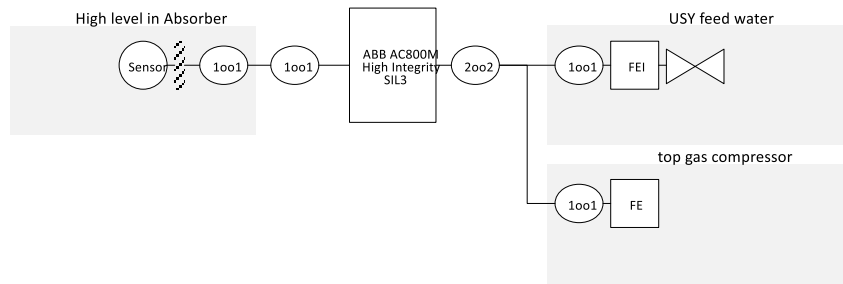
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-003 SIL1 High level (347LZ302) in Absorber (347YM100) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 15 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
2.16E-03	463.8	2	2	N/A	53.34



The SIF Conceptual Design is modeled as shown in the picture below.



4.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-003 SIL1 High level (347LZ302) in Absorber (347YM100) Safety Instrumented Function is quantified as follows.

Table 16 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.19E-03	2	N/A	263.56	0	2

Number of Sensor group(s): 1
Voting between groups: 1001
β-factor [%]: N/A

4.3.1.1 Sensor Group 1: High level in Absorber

The information and reliability data underneath describe the High level in Absorber sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Liquiphant M/S + FEL 58	2006.2.02	-	-	B	-
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off					
PR Electronics 9202, relay output	2015.2.03	-	-	B	2
ABB AC800M High Integrity SIL3 [DigitalIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 90		

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group High level in Absorber.

Table 17 Reliability Data Sensor Group High level in Absorber

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Liquiphant M/S + FEL 58	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	8.39E-08 <i>9.01E-08</i>	8.53E-08 <i>8.53E-08</i>	6.21E-09 <i>0.00E+00</i>	5.92E-08 <i>5.92E-08</i>	0.00E+00 <i>0.00E+00</i>	2.00E-09 <i>2.00E-09</i>	1.02E-07 <i>1.02E-07</i>
PR Electronics 9202, relay output	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>9.10E-08</i>	1.40E-07 <i>1.40E-07</i>	9.10E-08 <i>0.00E+00</i>	4.60E-08 <i>4.60E-08</i>	3.90E-08 <i>3.90E-08</i>	1.00E-09 <i>1.00E-09</i>	1.50E-07 <i>1.50E-07</i>
ABB AC800M High Integrity SIL3 [DigitalIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>2.38E-08</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>0.00E+00</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
								SFF [%]		81.9
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

4.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-003 SIL1 High level (347LZ302) in Absorber (347YM100) Safety Instrumented Function is quantified as follows.

Table 18 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.75E-08	3	N/A	70.96	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High
Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 19 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
						SFF [%]	100.0

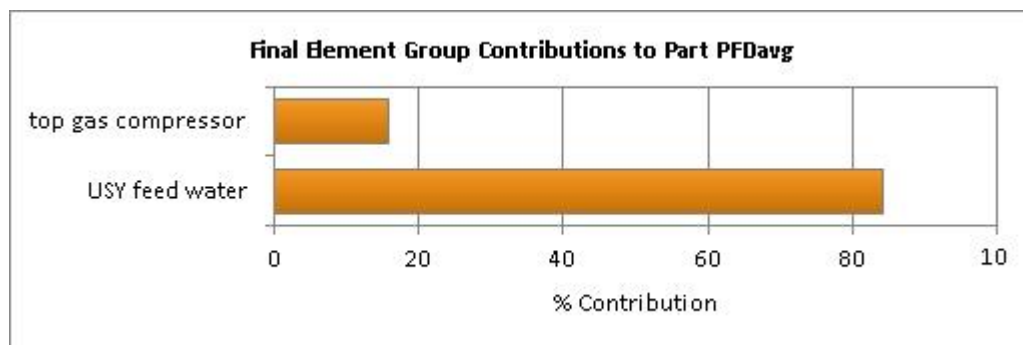
4.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-003 SIL1 High level (347LZ302) in Absorber (347YM100) Safety Instrumented Function is quantified as follows.

Table 20 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.68E-04	2	N/A	1162.96	0	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
β-factor [%]:



4.3.3.1 Final Element Group 1: USY feed water

The information and reliability data underneath describe the USY feed water final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Metso Norgren		-	-	A	3
Metso VPVL		-	-	A	3
Metso 7000 9000		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 85
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group USY feed water.

Table 21 Reliability Data Final Element Group USY feed water

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Metso Norgren		2.28E-09		2.28E-11			
Metso VPVL		4.25E-08		8.49E-09			
Metso 7000 9000		7.57E-08		4.84E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		79.8
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

4.3.3.2 Final Element Group 2: top gas compressor

The information and reliability data underneath describe the top gas compressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group top gas compressor.

Table 22 Reliability Data Final Element Group top gas compressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		89.5
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

5 SIF-004 SIL2 Low level (347LZ300) in Absorber (347YM100)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-004 SIL2 Low level (347LZ300) in Absorber (347YM100) Safety Instrumented Function.

5.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Low level (347LZ300) in Absorber (347YM100)
 SIF Tag: SIF-004
 SIF Description: SIL2 Low level (347LZ300) in Absorber (347YM100) stop Rich MEA to Stripper -> 347SV672 close and 347SV673 close

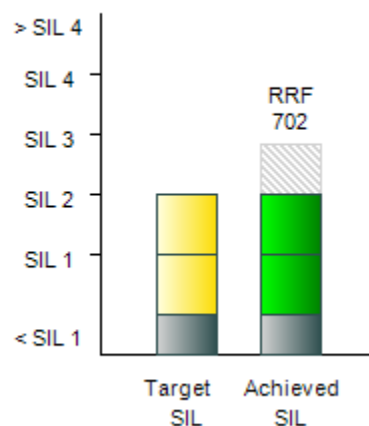
5.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 702



5.3 SIL Verification

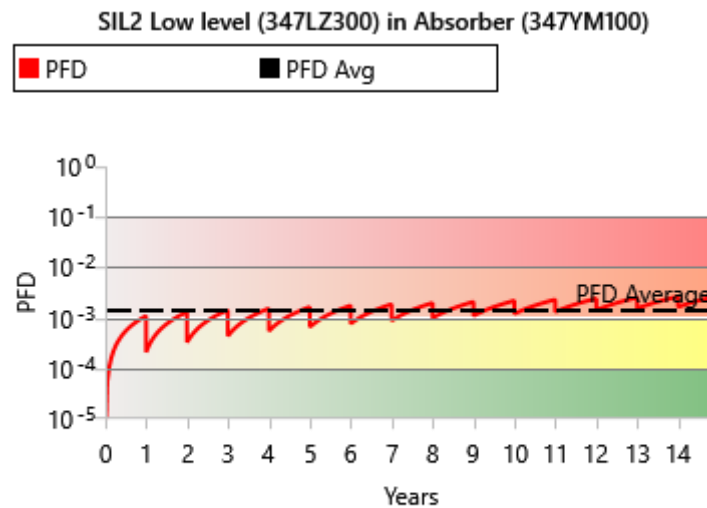
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-004 SIL2 Low level (347LZ300) in Absorber (347YM100) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
 Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst:
Design References
Comments:

Analysis Date: 10/4/2019

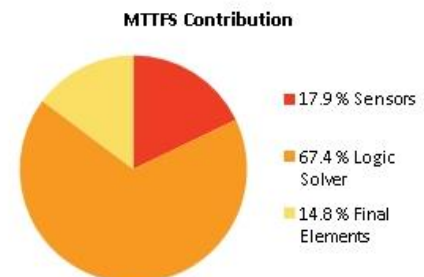
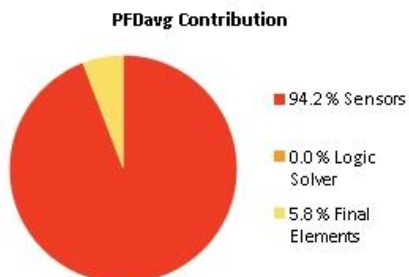


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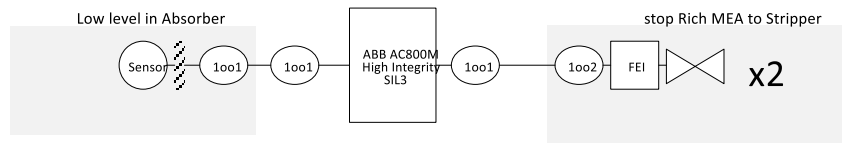
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-004 SIL2 Low level (347LZ300) in Absorber (347YM100) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 23 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.42E-03	702	2	2	N/A	48.67



The SIF Conceptual Design is modeled as shown in the picture below.



5.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-004 SIL2 Low level (347LZ300) in Absorber (347YM100) Safety Instrumented Function is quantified as follows.

Table 24 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.34E-03	2	N/A	272.46	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

5.3.1.1 Sensor Group 1: Low level in Absorber

The information and reliability data underneath describe the Low level in Absorber sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Liquiphant M/S + FEL 58	2006.2.02	-	-	B	-
Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
PR Electronics 9202, relay output	2015.2.03	-	-	B	2
ABB AC800M High Integrity SIL3 [DigitalIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Low level in Absorber.

Table 25 Reliability Data Sensor Group Low level in Absorber

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Liquiphant M/S + FEL 58	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	6.56E-08 <i>7.83E-08</i>	8.30E-08 <i>8.30E-08</i>	1.27E-08 <i>0.00E+00</i>	7.27E-08 <i>7.27E-08</i>	0.00E+00 <i>0.00E+00</i>	2.00E-09 <i>2.00E-09</i>	1.02E-07 <i>1.02E-07</i>
PR Electronics 9202, relay output	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>9.10E-08</i>	1.40E-07 <i>1.40E-07</i>	9.10E-08 <i>0.00E+00</i>	4.60E-08 <i>4.60E-08</i>	3.90E-08 <i>3.90E-08</i>	1.00E-09 <i>1.00E-09</i>	1.50E-07 <i>1.50E-07</i>
ABB AC800M High Integrity SIL3 [DigitalIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>2.38E-08</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>0.00E+00</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
								SFF [%]		79.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

5.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-004 SIL2 Low level (347LZ300) in Absorber (347YM100) Safety Instrumented Function is quantified as follows.

Table 26 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.90E-08	3	N/A	72.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High
Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 27 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
						SFF [%]	100.0

5.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-004 SIL2 Low level (347LZ300) in Absorber (347YM100) Safety Instrumented Function is quantified as follows.

Table 28 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
8.29E-05	3	N/A	329.69	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

5.3.3.1 Final Element Group 1: stop Rich MEA to Stripper

The information and reliability data underneath describe the stop Rich MEA to Stripper final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Metso Norgren		-	-	A	3
B1J		-	-	A	3
Metso 7000 9000		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group stop Rich MEA to Stripper.

Table 29 Reliability Data Final Element Group stop Rich MEA to Stripper

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Metso Norgren		2.28E-09		2.28E-11			
B1J				2.40E-08			
Metso 7000 9000		7.57E-08		4.84E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		74.0
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

6 SIF-005 SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101

This chapter details the SIL Verification results, selections, and assumptions for the SIF-005 SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101 Safety Instrumented Function.

6.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101
SIF Tag	SIF-005
SIF Description:	High level (347LS305 H1) filter separator 347SE101 stop 346PU101 (top gas compressor)

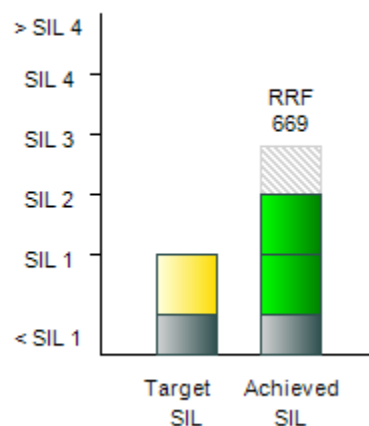
6.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 669.3



6.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-005 SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101 Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:

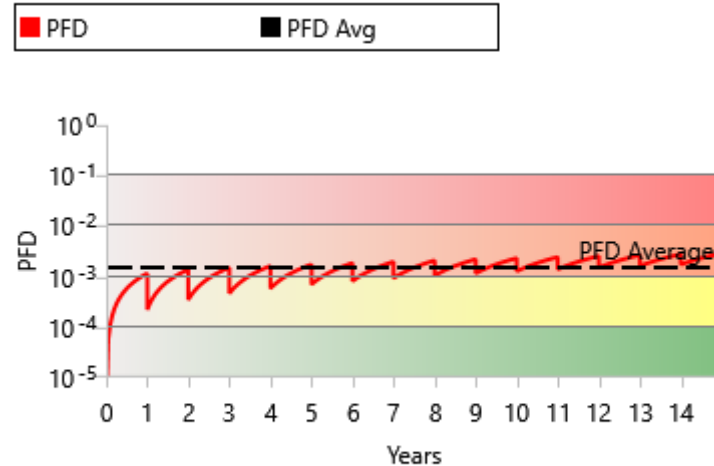
Analysis Date:

10/4/2019

Design References

Comments:

SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101

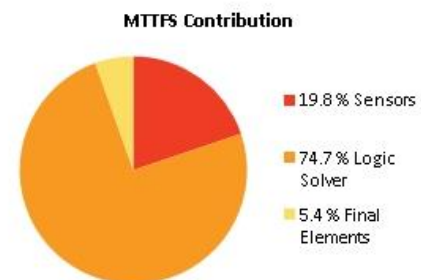
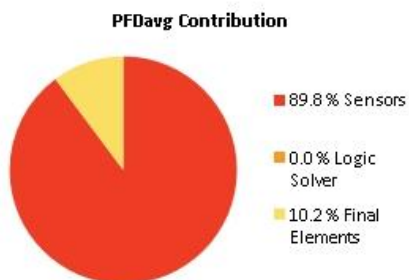


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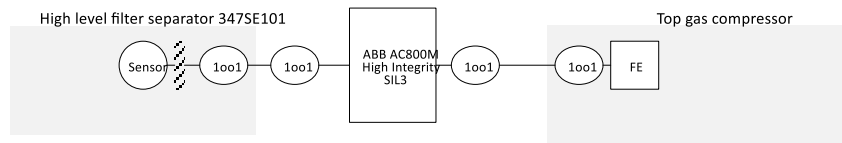
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-005 SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101 Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 30 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.49E-03	669.3	2	2	N/A	53.99



The SIF Conceptual Design is modeled as shown in the picture below.



6.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-005 SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101 Safety Instrumented Function is quantified as follows.

Table 31 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.34E-03	2	N/A	272.46	0	2

Number of Sensor group(s): 1
 Voting between groups: 1001
 β-factor [%]: N/A

6.3.1.1 Sensor Group 1: High level filter separator 347SE101

The information and reliability data underneath describe the High level filter separator 347SE101 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
 HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Liquiphant M/S + FEL 58	2006.2.02	-	-	B	-
Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
PR Electronics 9202, relay output	2015.2.03	-	-	B	2
ABB AC800M High Integrity SIL3 [DigitalIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group High level filter separator 347SE101.

Table 32 Reliability Data Sensor Group High level filter separator 347SE101

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Liquiphant M/S + FEL 58	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	6.56E-08 <i>7.83E-08</i>	8.30E-08 <i>8.30E-08</i>	1.27E-08 <i>0.00E+00</i>	7.27E-08 <i>7.27E-08</i>	0.00E+00 <i>0.00E+00</i>	2.00E-09 <i>2.00E-09</i>	1.02E-07 <i>1.02E-07</i>
PR Electronics 9202, relay output	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>9.10E-08</i>	1.40E-07 <i>1.40E-07</i>	9.10E-08 <i>0.00E+00</i>	4.60E-08 <i>4.60E-08</i>	3.90E-08 <i>3.90E-08</i>	1.00E-09 <i>1.00E-09</i>	1.50E-07 <i>1.50E-07</i>
ABB AC800M High Integrity SIL3 [Digitalln]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>2.38E-08</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>0.00E+00</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
								SFF [%]		79.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

6.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-005 SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101 Safety Instrumented Function is quantified as follows.

Table 33 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.90E-08	3	N/A	72.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High
Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 34 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
						SFF [%]	100.0

6.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-005 SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101 Safety Instrumented Function is quantified as follows.

Table 35 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.53E-04	2	N/A	990.72	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

6.3.3.1 Final Element Group 1: Top gas compressor

The information and reliability data underneath describe the Top gas compressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12		Proof Test Coverage [%]: 90			
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Top gas compressor.

Table 36 Reliability Data Final Element Group Top gas compressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		89.5
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

7 SIF-006 SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper

This chapter details the SIL Verification results, selections, and assumptions for the SIF-006 SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper Safety Instrumented Function.

7.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper
SIF Tag	SIF-006
SIF Description:	Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper -> 347SV672 close and 347SV673 close

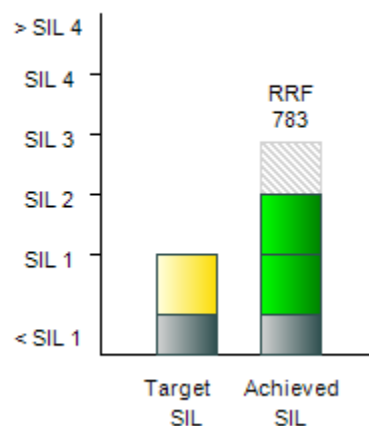
7.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 783.5



7.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-006 SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:

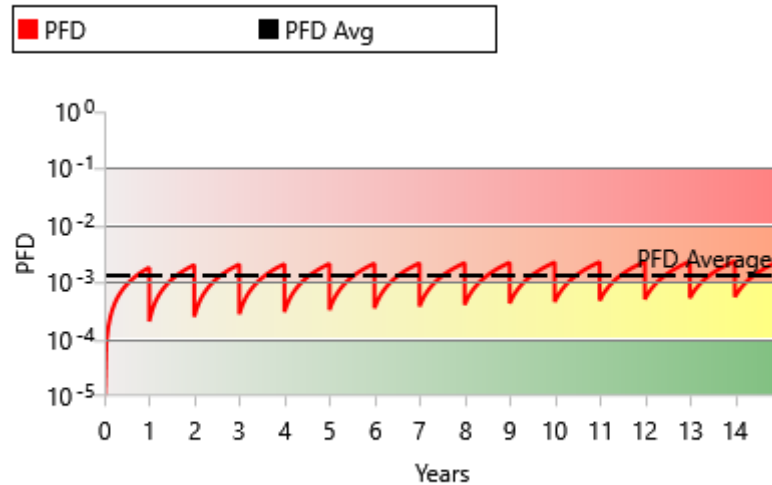
Analysis Date:

10/4/2019

Design References

Comments:

SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper



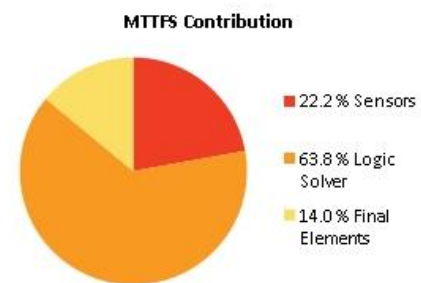
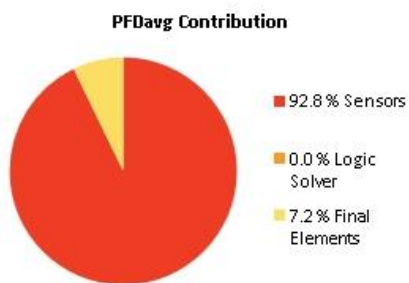
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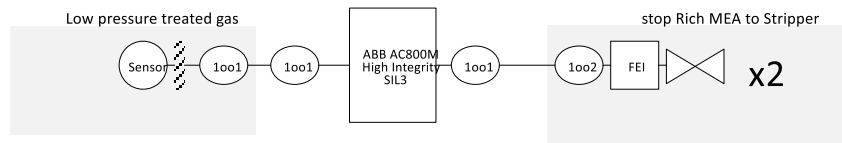
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-006 SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 37 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.28E-03	783.5	2	2	N/A	46.24



The SIF Conceptual Design is modeled as shown in the picture below.



7.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-006 SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper Safety Instrumented Function is quantified as follows.

Table 38 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.18E-03	2	N/A	208.68	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

7.3.1.1 Sensor Group 1: Low pressure treated gas

The information and reliability data underneath describe the Low pressure treated gas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2
Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
PR Electronics 9106, 1 active in/1 active out	2014.4.03	-	-	A	2
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Low pressure treated gas.

Table 39 Reliability Data Sensor Group Low pressure treated gas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
PR Electronics 9106, 1 active in/1 active out	1.46E-07 <i>0.00E+00</i>	2.70E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.46E-07</i>	0.00E+00 <i>0.00E+00</i>	4.10E-08 <i>6.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.77E-07 <i>1.77E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		73.4
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

7.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-006 SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper Safety Instrumented Function is quantified as follows.

Table 40 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.00E-08	3	N/A	72.45	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β-factor [%]: 1

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 41 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		100.0

7.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-006 SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper Safety Instrumented Function is quantified as follows.

Table 42 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.19E-05	3	N/A	329.71	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

7.3.3.1 Final Element Group 1: stop Rich MEA to Stripper

The information and reliability data underneath describe the stop Rich MEA to Stripper final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2H	PIU	ACTYPE	SIL CAP
Metso Norgren		-	-	A	3
B1J		-	-	A	3
Metso 7000 9000		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 88

Proof Test Execution:

Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group stop Rich MEA to Stripper.

Table 43 Reliability Data Final Element Group stop Rich MEA to Stripper

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Metso Norgren		2.28E-09		2.28E-11			
B1J				2.40E-08			
Metso 7000 9000		7.57E-08		4.84E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		74.0
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

8 SIF-007 SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps

This chapter details the SIL Verification results, selections, and assumptions for the SIF-007 SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps Safety Instrumented Function.

8.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps
SIF Tag	SIF-007
SIF Description:	High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps

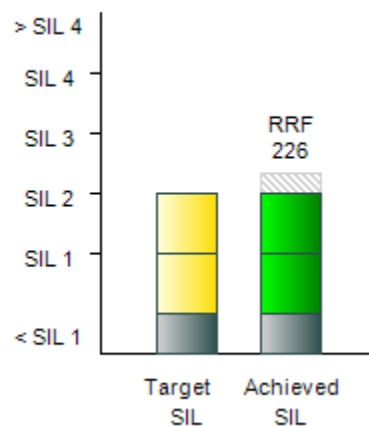
8.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 226.8



8.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-007 SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

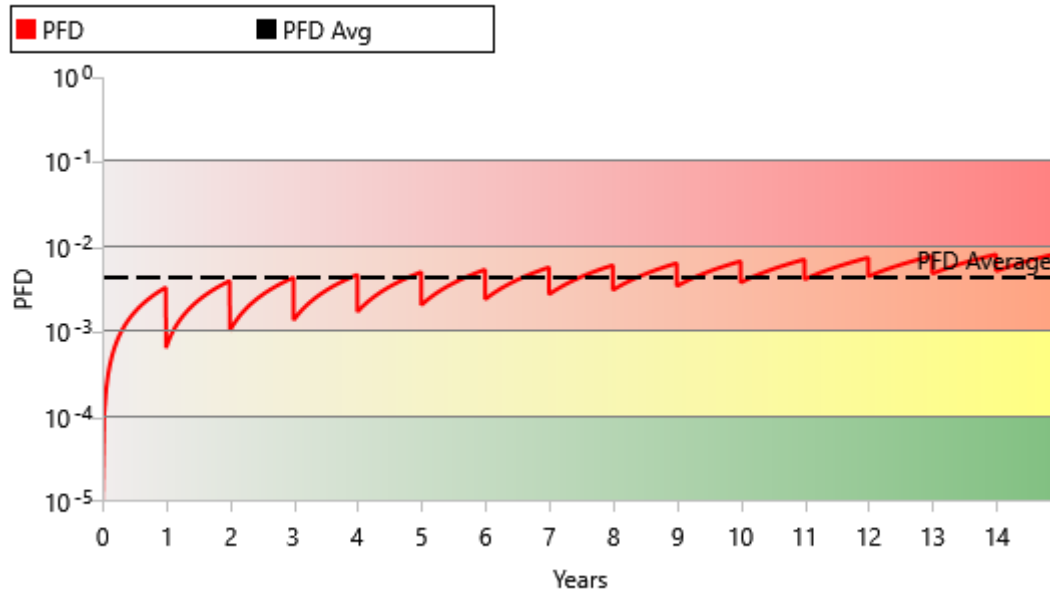
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

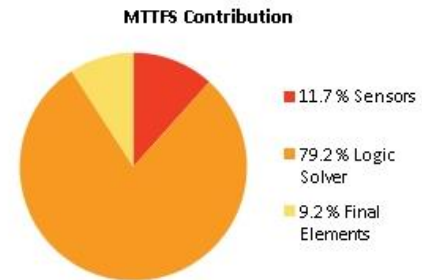
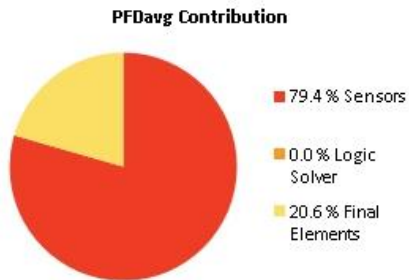
SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps



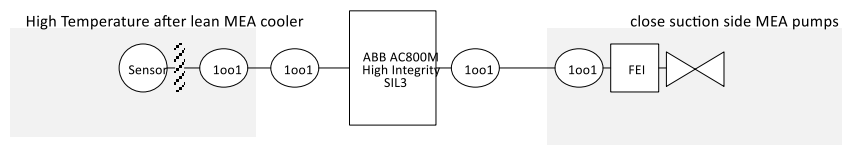
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-007 SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 44 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.41E-03	226.8	2	2	N/A	57.36



The SIF Conceptual Design is modeled as shown in the picture below.



8.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-007 SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps Safety Instrumented Function is quantified as follows.

Table 45 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.50E-03	2	N/A	491.7	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

8.3.1.1 Sensor Group 1: High Temperature after lean MEA cooler

The information and reliability data underneath describe the High Temperature after lean MEA cooler sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

SERH VERSION

2H

PIU

ACTYPE

SILCAP

Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off					
PR Electronics 9106, 1 active in/1 active out	2014.4.03	-	-	A	2
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group High Temperature after lean MEA cooler.

Table 46 Reliability Data Sensor Group High Temperature after lean MEA cooler

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.24E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
PR Electronics 9106, 1 active in/1 active out	1.46E-07 <i>0.00E+00</i>	2.70E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>2.70E-08</i>	0.00E+00 <i>0.00E+00</i>	4.10E-08 <i>1.87E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.77E-07 <i>1.77E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	43.0
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

8.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-007 SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps Safety Instrumented Function is quantified as follows.

Table 47 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.00E-08	3	N/A	72.45	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:	Architecture Type:	B		
HFT:	β -factor [%]:	1		
MRT [Hours]:				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	90	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 48 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	100.0

8.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-007 SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps Safety Instrumented Function is quantified as follows.

Table 49 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.07E-04	2	N/A	626.48	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

8.3.3.1 Final Element Group 1: close suction side MEA pumps

The information and reliability data underneath describe the close suction side MEA pumps final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Metso Norgren		-	-	A	3
B1J		-	-	A	3
Metso 7000 9000		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 88

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group close suction side MEA pumps.

Table 50 Reliability Data Final Element Group close suction side MEA pumps

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Metso Norgren		2.28E-09		2.28E-11			
B1J				2.40E-08			
Metso 7000 9000		7.57E-08		4.84E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	74.0
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

9 SIF-008 SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608

This chapter details the SIL Verification results, selections, and assumptions for the SIF-008 SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608 Safety Instrumented Function.

9.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608
 SIF Tag: SIF-008
 SIF Description: High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608

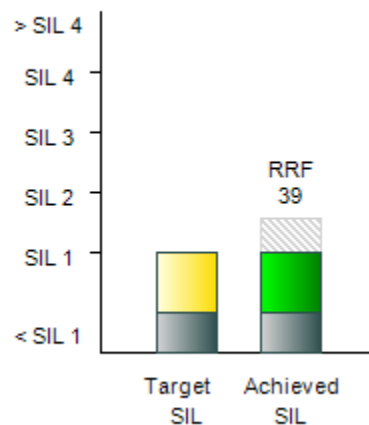
9.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 1 with RRF = 39.6



9.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-008 SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608 Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

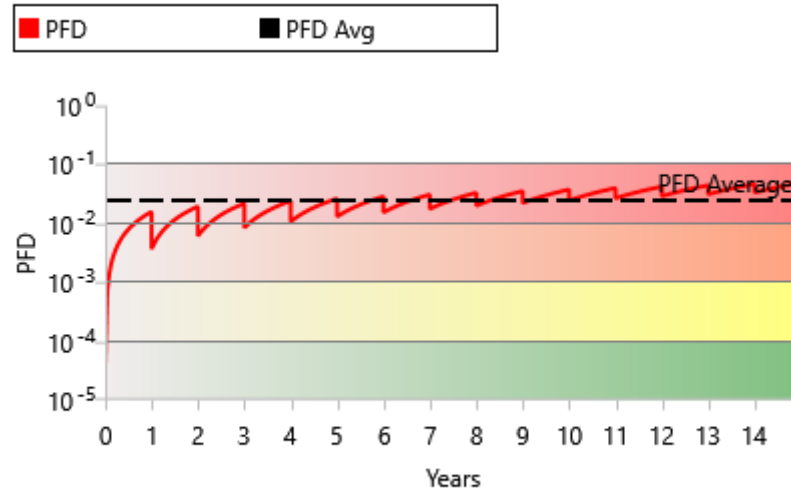
Mission Time: 15 years
 Startup time: 8 hours

The SIF operates in the Low demand mode.

Analyst:
Design References
Comments:

Analysis Date: 10/4/2019

SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608



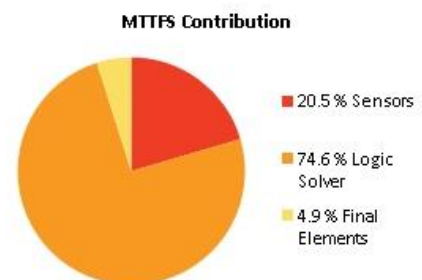
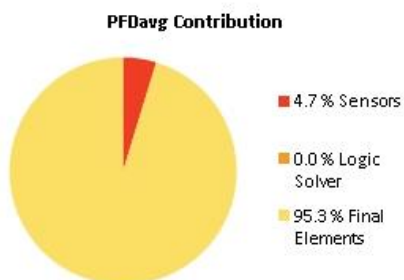
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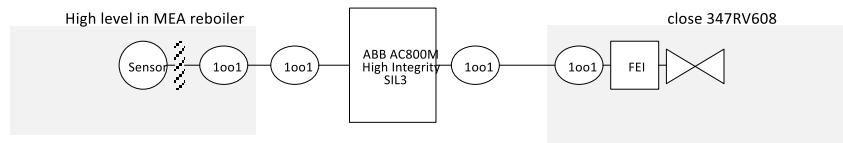
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-008 SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608 Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table S1 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
2.52E-02	39.6	1	2	N/A	53.9



The SIF Conceptual Design is modeled as shown in the picture below.



9.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-008 SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608 Safety Instrumented Function is quantified as follows.

Table 52 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.19E-03	2	N/A	263.56	0	2

Number of Sensor group(s): 1
Voting between groups: 1001
β-factor [%]: N/A

9.3.1.1 Sensor Group 1: High level in MEA reboiler

The information and reliability data underneath describe the High level in MEA reboiler sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Liquiphant M/S + FEL 58	2006.2.02	-	-	B	-
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
PR Electronics 9202, relay output	2015.2.03	-	-	B	2
ABB AC800M High Integrity SIL3 [DigitalIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group High level in MEA reboiler.

Table 53 Reliability Data Sensor Group High level in MEA reboiler

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Liquiphant M/S + FEL 58	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	8.39E-08 <i>9.01E-08</i>	8.53E-08 <i>8.53E-08</i>	6.21E-09 <i>0.00E+00</i>	5.92E-08 <i>5.92E-08</i>	0.00E+00 <i>0.00E+00</i>	2.00E-09 <i>2.00E-09</i>	1.02E-07 <i>1.02E-07</i>
PR Electronics 9202, relay output	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>9.10E-08</i>	1.40E-07 <i>1.40E-07</i>	9.10E-08 <i>0.00E+00</i>	4.60E-08 <i>4.60E-08</i>	3.90E-08 <i>3.90E-08</i>	1.00E-09 <i>1.00E-09</i>	1.50E-07 <i>1.50E-07</i>
ABB AC800M High Integrity SIL3 [Digitalln]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	7.19E-09 <i>2.38E-08</i>	3.45E-09 <i>3.45E-09</i>	1.66E-08 <i>0.00E+00</i>	1.66E-12 <i>1.66E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	9.23E-09 <i>9.23E-09</i>
									SFF [%]	81.9
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

9.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-008 SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608 Safety Instrumented Function is quantified as follows.

Table 54 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.77E-08	3	N/A	72.22	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High
Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	90	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 55 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital In Module: DI880/TU842/843	1.94E-07	4.80E-09	2.16E-07	1.15E-10	3.89E-08	3.93E-10	1.86E-07
Channel: Digital In	7.19E-09	3.45E-09	1.66E-08	1.66E-12			9.23E-09
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	100.0

9.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-008 SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608 Safety Instrumented Function is quantified as follows.

Table 56 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.40E-02	2	N/A	1096.4	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

9.3.3.1 Final Element Group 1: close 347RV608

The information and reliability data underneath describe the close 347RV608 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Metso Norgren		-	-	A	3
Metso VD		-	-	A	2
Metso GU		-	-	A	2
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 85
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group close 347RV608.

Table 57 Reliability Data Final Element Group close 347RV608

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Metso Norgren		2.28E-09		2.28E-11			
Metso VD				5.60E-07			
Metso GU				1.15E-06			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _h COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

10 SIF-009 SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber

This chapter details the SIL Verification results, selections, and assumptions for the SIF-009 SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber Safety Instrumented Function.

10.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber
SIF Tag	SIF-009
SIF Description:	Low differential pressure 347PD339 Pump discharge and absorber (347YM100) Close 347SV671 Lean MEA to Absorber Close 347RV667 lean MEA to Absorber

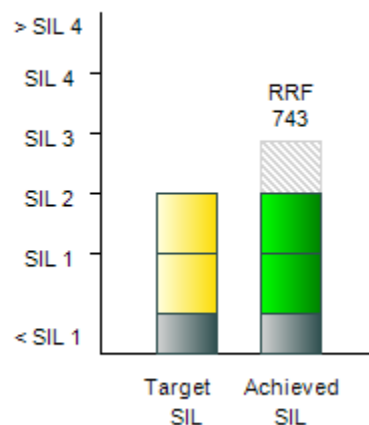
10.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 743.4



10.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-009 SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to

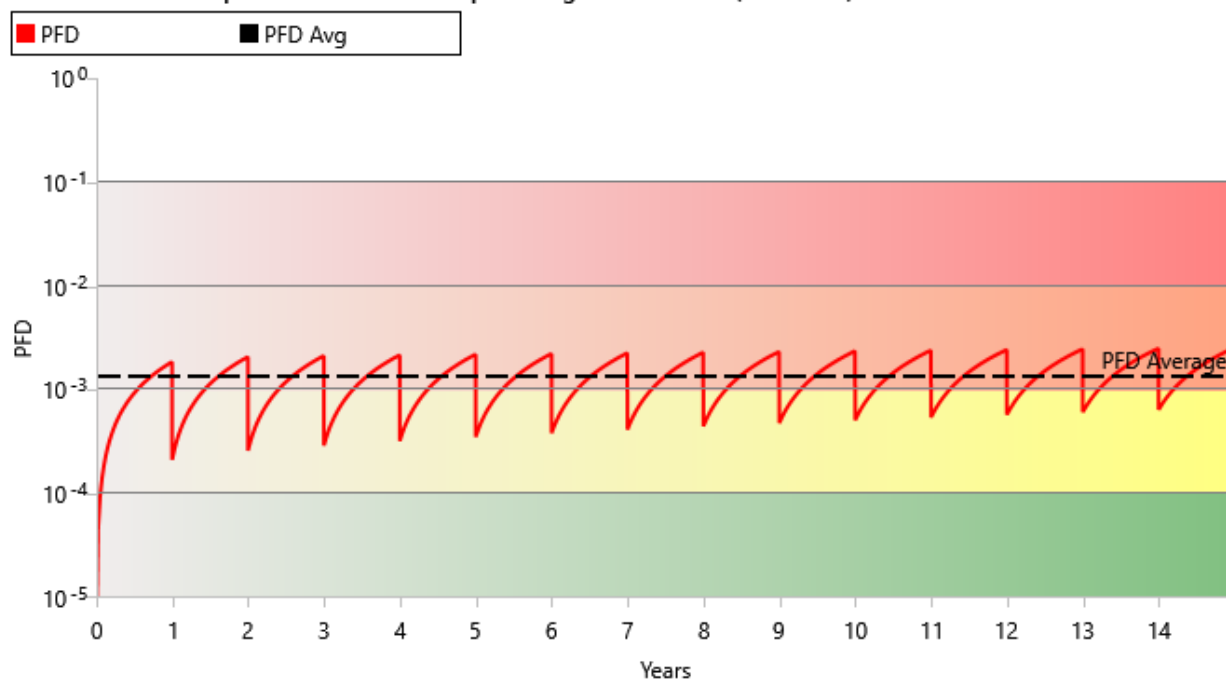
Absorber Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

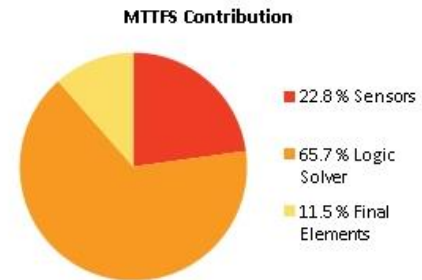
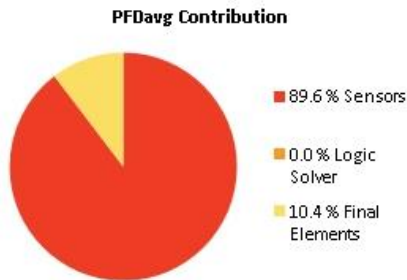
SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber



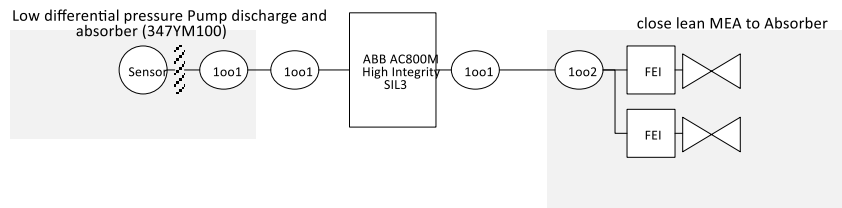
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-009 SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 58 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.35E-03	743.4	2	2	N/A	47.59



The SIF Conceptual Design is modeled as shown in the picture below.



10.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-009 SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber Safety Instrumented Function is quantified as follows.

Table 59 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.21E-03	2	N/A	208.69	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

10.3.1.1 Sensor Group 1: Low differential pressure Pump discharge and absorber (347YM100)

The information and reliability data underneath describe the Low differential pressure Pump discharge and absorber (347YM100) sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Deltabar S PMD75, FMD77, FMD78	2006.2.02	-	-	B	2

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

PR Electronics 9106, 1 active in/1 active out	2014.4.03	-	-	A	2
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Low differential pressure Pump discharge and absorber (347YM100).

Table 60 Reliability Data Sensor Group Low differential pressure Pump discharge and absorber (347YM100)

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Deltabar S PMD75, FMD77, FMD78	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.89E-08 <i>1.37E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	4.30E-07 <i>4.30E-07</i>
PR Electronics 9106, 1 active in/1 active out	1.46E-07 <i>0.00E+00</i>	2.70E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.46E-07</i>	0.00E+00 <i>0.00E+00</i>	4.10E-08 <i>6.80E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.77E-07 <i>1.77E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		73.1
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

10.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-009 SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber Safety Instrumented Function is quantified as follows.

Table 61 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.00E-08	3	N/A	72.45	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3
 SERH VERSION 2013.1.02
 PIU -
 SIL Cap N/A

Logic Solver Voting: Architecture Type: B
 HFT: 1 β -factor [%]: 1
 MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90
 Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 62 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	100.0

10.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-009 SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber Safety Instrumented Function is quantified as follows.

Table 63 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.40E-04	3	N/A	413.74	1	2

Number of Final Element group(s): 1
 Voting between groups: 1oo1
 β -factor [%]: N/A

10.3.3.1 Final Element Group 1: close lean MEA to Absorber

The information and reliability data underneath describe the close lean MEA to Absorber final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Diverse
HFT: 1 β -factor [%]: 10

Equipment Leg (1):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Metso Norgren		-	-	A	3
B1J		-	-	A	3
Metso 7000 9000		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

Equipment Leg (2):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Metso Norgren		-	-	A	3
Metso VD		-	-	A	2
Metso GU		-	-	A	2
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 85
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group close lean MEA to Absorber.

Table 64 Reliability Data Final Element Group close lean MEA to Absorber

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
347SV671							
Metso Norgren		2.28E-09		2.28E-11			
B1J				2.40E-08			
Metso 7000 9000		7.57E-08		4.84E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	74.0
						ROUTE 2 _H COMPLIANT	-
347RV667							
Metso Norgren		2.28E-09		2.28E-11			
Metso VD				5.60E-07			
Metso GU				1.15E-06			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	5.9
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

11 SILver™ Group Reuse Overview

This chapter provides an overview of the various groups that were defined during the SIL verification analyses of the Hybrit Pilot Plant project. The reuse of groups is available between Safety Instrumented Functions to account for identical hardware used in those different Safety Instrumented Functions. The group reuse allows users to specify a specific group once, reuse it, and with all future changes made to a single group cascading to all Safety Instrumented Functions it is used in. It is instrumental for the user to review all SIFs a single group is used in when changes are made to that group.

This overview indicates the groups that were reused, the number of times they were reused, and the Safety Instrumented Functions that they were used in.

11.1 Sensor Groups Reused

Gasttryck toppgas (used 4 times)	SIF-013, SIF-014, SIF-011, SIF-012
Gasttryck transferline (used 3 times)	SIF-021, SIF-024, SIF-020
Tryckvakt O2-inj (used 3 times)	SIF-022, SIF-023, SIF-020
Temperatur konvektionsdel (used 2 times)	SIF-027, SIF-028

11.2 Logic Solver Groups Reused

ABB AC800M High Integrity SIL3 (used 72 times)	SIF-996, SIF-997, SIF-998, SIF-026, SIF-001, SIF-034, SIF-002, SIF-003, SIF-004, SIF-005, SIF-051, SIF-052, SIF-053, SIF-054, SIF-056, SIF-058, SIF-059, SIF-060, SIF-061, SIF-062, SIF-063, SIF-076, SIF-077, SIF-064, SIF-065, SIF-067, SIF-068, SIF-069, SIF-070, SIF-071, SIF-074, SIF-073, SIF-072, SIF-075, SIF-019, SIF-013, SIF-014, SIF-017, SIF-018, SIF-081, SIF-039, SIF-040, SIF-042, SIF-045, SIF-046, SIF-048, SIF-011, SIF-021, SIF-024, SIF-022, SIF-023, SIF-020, SIF-025, SIF-027, SIF-028, SIF-029, SIF-015, SIF-012, SIF-031, SIF-032, SIF-033, SIF-035, SIF-036, SIF-037, SIF-038, SIF-041, SIF-043, SIF-044, SIF-047, SIF-080, SIF-079, SIF-082
ABB AC800M High Integrity SIL3 (used 9 times)	SIF-006, SIF-007, SIF-008, SIF-009, SIF-049, SIF-050, SIF-055, SIF-057, SIF-066

11.3 Final Element Groups Reused

Avstängning O2 (used 6 times)	SIF-026, SIF-037, SIF-043, SIF-044, SIF-079, SIF-082
Trip red.gaskompressor (used 6 times)	SIF-050, SIF-025, SIF-043, SIF-044, SIF-047, SIF-038
Trip proc.gaskompressor (used 5 times)	SIF-055, SIF-042, SIF-045, SIF-046, SIF-048
Trip red.gaskompressor (used 2 times)	SIF-049, SIF-041
Avstängningsventil 2 överhettad ånga till PGH (used 8 times)	SIF-076, SIF-016, SIF-017, SIF-018, SIF-028, SIF-041, SIF-047, SIF-080
H2 till brännare PGH (used 2 times)	SIF-019, SIF-079
NG till brännare PGH (used 2 times)	SIF-019, SIF-082
Close NG till kylkrets	SIF-042, SIF-045, SIF-046, SIF-048

(used 4 times)

Avstängning O2 PT12

(used 18 times)

NG till brännare PGH PT12

(used 5 times)

Avstängningsventil 2 överhettad ånga
till PGH PT 6

(used 5 times)

SIF-021, SIF-022, SIF-024, SIF-023, SIF-020, SIF-025, SIF-027, SIF-029,
SIF-015, SIF-031, SIF-032, SIF-033, SIF-035, SIF-036, SIF-041, SIF-047,
SIF-081, SIF-038

SIF-025, SIF-027, SIF-029, SIF-031, SIF-032

SIF-012, SIF-043, SIF-044, SIF-079, SIF-082

12 Abbreviations and Definitions

12.1 Abbreviations

HFT	Hardware Fault Tolerance
IPL	Independent Protection Layers
MoonN	"M" out of "N"
MTTFS	Mean Time To Fail Spurious
MRT	Mean Repair Time
PDF	Probability of Failure on Demand
PDFavg	Average Probability of Failure on Demand
PFH	Probability of a Dangerous Failure per Hour
PIU	Proven In Use
PTC	Proof Test Coverage
PTI	Proof Test Interval
RRF	Risk Reduction Factor
SERH	Safety Equipment Reliability Handbook
SFF	Safe Failure Fraction
SIF	Safety Instrumented Function
SIL	Safety Integrity Level
SIS	Safety Instrumented System
SSI	Site Safety Index
SRS	Safety Requirements Specification
β -factor	Beta factor, indicating common cause susceptibility
SD	Safe Detected
SU	Safe Undetected
DD	Dangerous Detected
DU	Dangerous Undetected
AD	Annunciation Detected
AU	Annunciation Undetected
NE	No Effect

12.2 Definitions

Architectural Constraints	Indication of maximum SIL achievable of a SIF's subsystems architecture based on minimum hardware fault tolerance requirements. This is determined solely on the basis of Type A or Type B device selection, redundancy (hardware fault tolerance), and the safe failure fraction (calculated or conservatively assumed if no data is provided). It does not pertain to Systematic Capability or certification levels. Architectural Constraint limits are determined based on IEC 61508 or IEC 61511 tables.
Mission Time	This is the time period that the SIF is expected to be operational. Alternatively this is the time period between points in time where the SIF can be considered as new. Note that this is not the proof test interval because of imperfect proof tests considerations. The mission time could, for example, correspond to the major turnaround period of a unit.
Startup Time	This represents the time it will take to perform maintenance and repair on a unit after a spurious trip plus the time it will take to restart the unit. This is typically longer than just the average repair time.

Systematic Capability

Indication of the maximum SIL achievable based on the SIF's equipment items systematic integrity alone. This is determined on the basis of the IEC 61508 certification level of each equipment item and/or the systematic integrity claim made by the user for a specific equipment item as part of its Proven In use justification. This addresses the requirement in IEC 61511 that all equipment used in a SIF must either be assessed per IEC 61508 or proven in use.

13 Disclaimer, Assumptions, Equipment Data

13.1 Disclaimer

The user of the exSILentia® software is responsible for verification of all results obtained and their applicability to any particular situation. Calculations are performed per guidelines in applicable international standards. *exida.com L.L.C.* accepts no responsibility for the correctness of the regulations or standards on which the tool is based. In particular, *exida.com L.L.C.* accepts no liability for decisions based on the results of this software. The *exida.com L.L.C.* guarantee is restricted to the correction of errors or deficiencies within a reasonable period when such errors or deficiencies are brought to our attention in writing. *exida.com L.L.C.* accepts no responsibility for adjustments made to this automatically generated report made by the user.

13.2 Assumptions SILver™

SIL Capability

If the user disables the **Consider IEC SIL Capability** option, it is assumed that the justification of equipment used is performed outside of the SILver tool as in that case the results of the SIL verification only address the quantitative requirements of IEC 61511.

A list of all other assumptions on which the SILver™ tool is based can be found in the exSILentia® user guide.

13.3 Equipment Data

exida has compiled a proprietary equipment failure database. This database is a compilation of failure data collected from a variety of public and confidential sources and presents an industry average. The database is published as the "Safety Equipment Reliability Handbook" ISBN-13: 978-1-934977-15-6. The reliability data collection process as described in this book applies to the SILver equipment data collection process.

The user is responsible for determining the applicability of the failure data to any particular environment. The stress levels assumed to determine the equipment failure rate are average worst case for an industrial environment. Descriptions of the environmental profiles used are included in the Safety Equipment Reliability Handbook. Accurate plant specific data is preferable to general industry average data. Industrial plant sites with high levels of stress must use failure rate data that is adjusted to a higher value to account for the specific conditions of the plant.